

PROJECT REPORTS

AI Learning & Study Assistant

A Project Report Submitted in Partial Fulfilment
of the Requirements for the Academic Project

POOJASRI N

Department of Computer Science and Engineering

LATHA MATHAVAN ENGINEERING COLLEGE

Academic Year: 2026-2027

TABLE OF CONTENTS

1. Project Overview
2. Objectives & Proposed Solution
3. Implementation & Results
4. Conclusion & Future Scope
5. References

Full Project: AI Learning & Study Assistant

Project Title

AI-Powered Learning and Study Assistant using Agentic AI



1. Problem Statement

Students often have difficulty creating study schedules, finding information from large study materials, revising topics, and preparing for examinations.

The proposed system is an **AI-powered Learning and Study Assistant** that acts as a personal study agent. It can understand the student's request, retrieve information from study materials, create personalized study plans, generate quizzes, and remember relevant student preferences.

2. Main Objective

The main objective is to develop an **Agentic AI Study Assistant** that helps students:

- Understand academic topics
 - Search information from uploaded study materials
 - Create personalized study plans
 - Generate practice questions and quizzes
 - Track previous interactions
 - Provide context-aware responses
-

3. What Our AI Agent Will Do

The student interacts with one chatbot.

Example 1 — Question Answering

Student:

Explain normalization in DBMS.

AI:

Explains normalization using information retrieved from the student's DBMS study material.

Example 2 — Study Plan

Student:

I have 5 days to prepare for DBMS.

AI:

Creates a 5-day study plan based on the subjects/topics.

Example 3 — Quiz

Student:

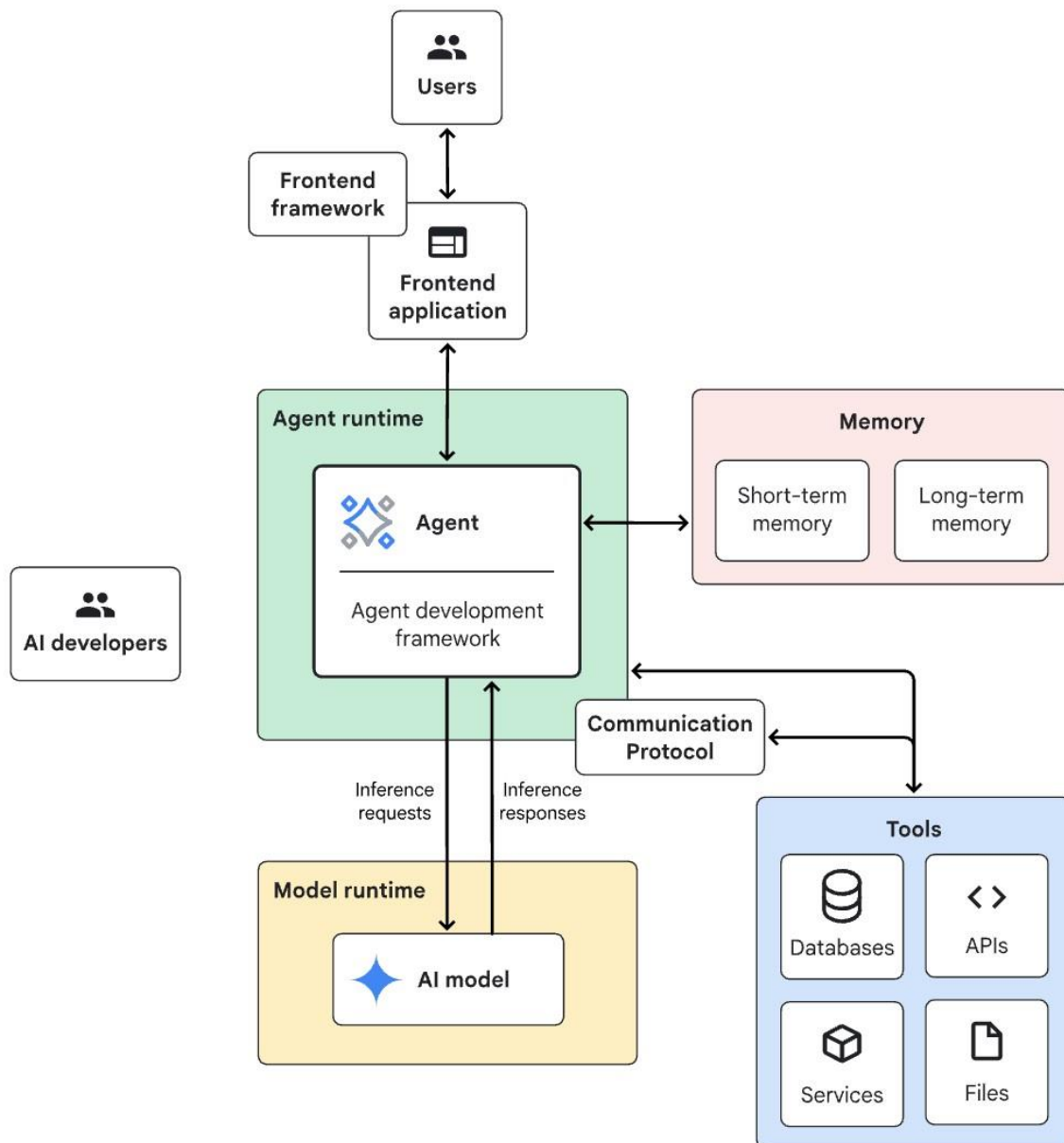
Give me a quiz on normalization.

AI:

Generates multiple-choice questions for practice.

4. Agentic AI Architecture

Our project will contain:



5. Three Important Agent Capabilities

A. RAG — Retrieval Augmented Generation

RAG allows the AI to answer questions using **your uploaded study materials**.

Flow:

```
PDF / Notes
  ↓
Document Processing
  ↓
Text Extraction
  ↓
Text Chunking
  ↓
Embeddings
  ↓
Vector Database
  ↓
Student Question
  ↓
Relevant Information Retrieved
  ↓
AI Generates Answer
```

So the AI doesn't have to depend only on general knowledge.

B. Memory

Memory allows the assistant to retain useful information from previous interactions.

For example:

```
Student Name: Student
Current Subject: DBMS
Exam: Semester Exam
Preferred Learning Style: Simple Explanation
```

The assistant can use this information in future responses.

C. Tools

The agent can decide when to use different tools.

Tools we can implement:

1. Study Plan Generator

Input:

```
Subject = DBMS  
Days = 5  
Study Hours = 3
```

Output:

```
Day 1 → ER Model  
Day 2 → Normalization  
Day 3 → SQL  
Day 4 → Transactions  
Day 5 → Revision + Quiz
```

2. Quiz Generator

Creates MCQs based on a selected topic.

3. Study Material Search

Searches the uploaded notes/PDFs.

6.Complete Project Modules

Our final project can have these modules:

Module 1 — User Interface

Student interacts with the AI assistant.

Module 2 — AI Agent

Understands the student's request and decides what action is required.

Module 3 — RAG System

Retrieves information from uploaded study materials.

Module 4 — Memory System

Stores useful conversation/student context.

Module 5 — Study Plan Tool

Creates personalized study schedules.

Module 6 — Quiz Tool

Generates questions for revision.

Module 7 — Document Processing

Processes PDFs and notes.

Module 8 — Response Generation

Produces the final answer in a student-friendly format.

7. Final Demo Scenario

For your final presentation, you can demonstrate something like:

Step 1: Upload a DBMS PDF.

Step 2:

Explain normalization.

AI retrieves the relevant information from the PDF.

Step 3:

Create a 5-day study plan for DBMS.

AI uses the study-plan tool.

Step 4:

Give me 5 MCQs on normalization.

AI uses the quiz-generation capability.

Step 5:

What topic was I studying earlier?

AI uses memory.

This single demo shows **RAG + Tools + Memory + Agent** — exactly the capabilities listed for the 5th use case.
